

Minimodule 3. Induction

List of exercises (from 7th edition of the book)

Section 5.1 Exercises 3, 9, 19, 31

3. Let $P(n)$ be the statement that $1^2 + 2^2 + \cdots + n^2 = n(n+1)(2n+1)/6$ for the positive integer n .

- What is the statement $P(1)$?
- Show that $P(1)$ is true, completing the basis step of the proof.
- What is the inductive hypothesis?
- What do you need to prove in the inductive step?
- Complete the inductive step, identifying where you use the inductive hypothesis.
- Explain why these steps show that this formula is true whenever n is a positive integer.

9. a) Find a formula for the sum of the first n even positive integers.

b) Prove the formula that you conjectured in part (a).

19. Let $P(n)$ be the statement that

$$1 + \frac{1}{4} + \frac{1}{9} + \cdots + \frac{1}{n^2} < 2 - \frac{1}{n},$$

where n is an integer greater than 1.

- What is the statement $P(2)$?
- Show that $P(2)$ is true, completing the basis step of the proof.
- What is the inductive hypothesis?
- What do you need to prove in the inductive step?
- Complete the inductive step.
- Explain why these steps show that this inequality is true whenever n is an integer greater than 1.

31. Prove that 2 divides $n^2 + n$ whenever n is a positive integer.

For more practice, you may also consider the following exercises 32 - 37, which are similar but require some other rearrangements to apply the inductive hypothesis.